ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Exploring Mixtures and Their Separation** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(b)** 50 per cent	30/60.
3	**(a)** only one component sublimes	Different behaviour on heating.
4	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
5	**(b)** 32 g	Water 100 g holds 32 g; crystals 64 - 32.
6	**(c)** iron and sulphur	Only iron is magnetic.
7	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
8	**(c)** 18 g	50 - 32.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Sand stays on the filter paper while the salt solution passes through; the difference in solubility explains the method.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true statements, but R does not explain why a fractionating column is needed in A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Mass of solution = 240 g (1); 24/240 x 100 = 10 per cent (1).	1+1
12	Boiling point is fixed (1); Heat supplies latent heat (1).	1+1
13	Slow cooling allows ordered growth (1); Few nuclei form (1).	1+1
14	Solute may have low solubility (1); e.g. a saturated solution of chalk or CO2 in water is dilute (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	320 g contains 120 g salt and 200 g water (1); At 20 degrees Celsius 200 g water holds 50 g (1); Crystals = 120 - 50 = 70 g (1).	1+1+1
16	A: 5/55 = 9.1 per cent (1); B: 8/108 = 7.4 per cent (1); A is more concentrated (1).	1+1+1
17	Salt gives ions that neutralise the charge on particles (1); Particles aggregate (1); Soap separates (1).	1+1+1
	OR: X = 0.27 (1); Y = 0.6, Z = 0.8 (1); X is most strongly adsorbed (1).	1+1+1
18	Ink spots from document and suspects (1); Run chromatography (1); Compare bands and R_f values (1).	1+1+1
	OR: Measure boiling point 100 degrees Celsius (1); Evaporate: no residue (1); Test conductivity (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Solute: salt; solvent: water (1). (ii) $20 + 180 = 200$ g (1). (iii) $\frac{20}{200} \times 100 = 10\%$ (1). (iv) For 5 per cent, total mass $= \frac{20}{0.05} = 400$ g (1); water to be added $= 400 - 200 = 200$ g (1).	1+1+1+2
20	Spot unknown and known dyes on the same paper (1); Run in a suitable solvent (1); Compare R_f values (1); Same R_f means same dye (1); Solvent changes R_f (1).	1+1+1+1+1
	OR: Screening and sedimentation (1); Alum coagulation and filtration (1); Chlorination for bacteria (1); Desalination (distillation or reverse osmosis) for dissolved salts (1); Testing quality (1).	1+1+1+1+1

Mistakes to avoid:

- Calculating mass per cent using the mass of the *solvent* instead of the mass of the *solution* (solute + solvent).
- Saying that a clear liquid must be a pure substance; a clear solution such as salt water is a mixture.
- Using filtration to separate a dissolved solid; filtration works only for insoluble particles, not for a solute.

- Confusing evaporation (solvent lost, solid left) with distillation (solvent collected by condensing); and using fractional distillation when the boiling points differ by more than 25 K.
- Thinking that the solubility of a gas also increases when the temperature rises; the solubility of gases in water decreases.
- Forgetting that a colloid does not settle and cannot be filtered, but still shows the Tyndall effect.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Exploring Mixtures and Their Separation** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** the one with R_f 0.9	Largest R_f.
2	**(d)** distillation	Boiling point.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
4	**(c)** 5 per cent (m/v)	20 g in 400 mL.
5	**(c)** dissolved impurities	Boiling point elevation.
6	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
7	**(b)** 500 mL	30/0.06.
8	**(d)** solvent extraction	Selective solubility.
9	**(c)** A is true but R is false.	Suspension particles are large and settle on standing, so R is false.
10	**(d)** A is false but R is true.	Particles of a true solution are too small to scatter light, so A is false; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Molecules pack into gaps (1); Interactions between molecules (1).	1+1
12	They form distinct layers (1); Miscible liquids form one layer (1).	1+1
13	12/100 x 750 (1); = 90 mL (1).	1+1
14	Suspension (1); Settles and is filtered (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Add water and stir; sugar dissolves (1); Filter: sand and chalk remain (1); Evaporate for sugar; separate chalk from sand by decantation or floatation (1).	1+1+1
16	Solubility depends on the solute (1); Different adsorption on paper (1); R_f varies (1).	1+1+1
17	Pure compound moves as one band (1); Mixture components have different solubilities and attractions to paper (1); Black ink separates into several coloured bands (1).	1+1+1
	OR: Vapour contains both (1); Close boiling points (1); Fractional distillation needed (1).	1+1+1
18	Sublimation: solid to vapour; evaporation: liquid to vapour (1); Camphor vs water (1); Needs different conditions (1).	1+1+1
	OR: Crush and filter juice (1); Evaporate to syrup (1); Crystallise and centrifuge crystals (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Evaporation (1). (ii) Salt is dissolved in the water as very small particles and passes through the filter paper (1). (iii) Crystallisation (recrystallisation) (1). (iv) Sunlight is free and gives slow evaporation, so crystals of salt form (1); boiling to dryness would waste fuel and leave the impurities mixed with the salt (1).	1+1+1+2
20	Petroleum gas/naphtha, petrol (1); Kerosene (1); Diesel (1); Lubricating oil, bitumen (1); Column gives a temperature gradient so fractions condense at levels (1).	1+1+1+1+1
	OR: Heat both in water bath (1); Record temperature every minute (1); Pure naphthalene stays constant at 80 degrees Celsius while melting (1); Mixture melts over a range at lower temperature (1); Plot cooling curve (1).	1+1+1+1+1

Mistakes to avoid:

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- Saying that a clear liquid must be a pure substance; a clear solution such as salt water is a mixture.
- Using filtration to separate a dissolved solid; filtration works only for insoluble particles, not for a solute.
- Confusing evaporation (solvent lost, solid left) with distillation (solvent collected by condensing); and using fractional distillation when the boiling points differ by more than 25 K.

- Thinking that the solubility of a gas also increases when the temperature rises; the solubility of gases in water decreases.
- Forgetting that a colloid does not settle and cannot be filtered, but still shows the Tyndall effect.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Exploring Mixtures and Their Separation** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** it keeps the jacket full for efficient cooling	Complete cooling.
2	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
3	**(a)** increased four times	12/3.
4	**(d)** to prevent bumping and give smooth boiling	Smooth boiling.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(c)** an emulsion	e.g. milk.
7	**(c)** the second component	Ascending order.
8	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
9	**(a)** Both A and R are true and R is the correct explanation of A.	The more soluble dye moves farther up the paper, which is what separates the dyes.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both statements are true; the percentage of nitrogen does not explain why air is homogeneous.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Heating forms iron sulphide compound (1); A magnet separates physically (1).	1+1
12	Sample dissolves in solvent in the container (1); Separation does not occur (1).	1+1
13	Vinegar particles are molecular in size (1); Soap forms micelles, larger aggregates (1).	1+1
14	Alum coagulates the suspended particles (1); They settle down faster (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sugar: crystallisation of concentrated juice (1); Oxygen: fractional distillation of liquid air (1); Boiling and solubility differences (1).	1+1+1
16	Alcohol with lower b.p. distils first (1); Composition of the remaining liquid changes (1); Final temperature approaches 100 (1).	1+1+1
17	Curve shows solubility vs temperature (1); Difference at two temperatures gives crystals (1); Choose solvents with steep curves (1).	1+1+1
	OR: Alum coagulates particles (1); They settle in clumps (1); Clear water decanted (1).	1+1+1
18	Atoms of carbon disturb the regular arrangement (1); Layers cannot slide easily (1); Hence harder (1).	1+1+1
	OR: Powdered solid dissolves faster (1); Higher temperature increases rate (1); Change only one factor (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Centrifugation (1). (ii) Cream (1). (iii) Any one: separating blood cells from plasma, separating butter from curd or cream, drying clothes in a washing machine (1). (iv) The spinning produces a strong outward force (1); denser particles are thrown to the bottom or outside while lighter particles stay near the centre (1).	1+1+1+2
20	Add salt until some stays undissolved (1); Stir, keep at the temperature (1); Weigh an evaporating dish (1); Evaporate a known mass of solution (1); Solubility = mass of salt/mass of water x 100 (1).	1+1+1+1+1
	OR: Medicine: purification of drugs by crystallisation (1); Food: sugar, salt, milk products (1); Mining: magnetic separation, froth floatation (1); Environment: waste water treatment (1); Oil spill clean-up (1).	1+1+1+1+1

Mistakes to avoid:

- Calculating mass per cent using the mass of the *solvent* instead of the mass of the *solution* (solute + solvent).

- Saying that a clear liquid must be a pure substance; a clear solution such as salt water is a mixture.
- Using filtration to separate a dissolved solid; filtration works only for insoluble particles, not for a solute.
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